

NODE CALIBRATION

Smart Signage System

Department of Software Development, I2ST.

Abstract

The management of distributed hardware nodes in modern smart signage systems demands a unified, efficient, and secure configuration platform. This project presents a comprehensive Windows Desktop Application, developed in C# using the .NET framework, designed to centrally manage and calibrate ESP32-based controller units deployed across multiple signage panels. The application integrates a local SQL Server database for persistent storage of node records, user credentials, and panel location descriptions, combined with a JSON-based settings file for application-level preferences.

The system provides a role-based authentication mechanism, ensuring that only authorized personnel can access critical functions. A dynamic user interface allows operators to define network nodes with unique parameters including IP addresses, panel serial numbers, and physical locations. Two distinct communication channels—direct USB serial and Over-the-Air HTTP—enable the deployment of configuration payloads to remote devices without the need for physical access. The application enforces data integrity through strict validation rules, preventing duplicate IP assignments and ensuring each hardware unit is calibrated only once. An integrated serial monitor and OTA management panel offer real-time feedback on device status and configuration success.

The ESP32 firmware, implemented in the Arduino framework, incorporates Ethernet connectivity, an HTTP server, MQTT telemetry, and persistent storage via the Preferences library. On receipt of a configuration, the device dynamically updates its network settings, sensor calibration constants, and operational parameters, then automatically restarts to apply changes. A factory reset capability allows remote recovery of misconfigured devices.

This solution significantly reduces the time and complexity involved in provisioning and maintaining a large network of signage controllers. It eliminates manual data entry through planned Excel import/export functionality and lays the groundwork for future integration of Modbus TCP-based diagnostic testing. The project demonstrates a robust, scalable, and practical approach to industrial device management, blending desktop application engineering with embedded systems communication.

CONTENTS

Chapter No.	Title	Page No.
Chapter 1	Introduction.	
1.1	Background.	
1.2	Proposed System.	
1.3	Scope and Objectives.	
Chapter 2	Requirement specification.	
Chapter 3	System Design.	
3.1	System Architecture.	
3.2	Flow Diagrams.	
3.3	Database Schema.	
Chapter 4	System Implementation.	
4.1	Main Window Interface.	
4.1.1	Menu Strip (Top Navigation Bar).	
4.1.2	Left Navigation Panel.	
4.2	Authentication Module.	
4.3	Node Management Module.	
4.4	Area Management Module.	
4.5	Panel Settings Module.	
4.6	Settings Module.	
4.7	Network Configuration Module.	
4.8	I/O Activity Module.	
4.9	Upload/Download Module.	
4.10	Tests Module.	